Secure By Design

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COURSE 1.6 / COMPANION WORKBOOK

See everything: the logging pipeline

Collect, protect & analyze before the fleet arrives

Build the model's collect → protect → analyze chain: Windows Event Forwarding and router syslog into SBD-SYSLOG01, hashed archives under a no-delete rule on SBD-AUDIT01, and working searches with one owned alert on SBD-SPLUNK01 — proven by a single event traced end to end.

MATERIALS EDITION 1.3 / COURSE 1.6

Updated 8 October 2026 · sbd-logging-pipeline

NAME _____ / DATE _____

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Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

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Before you start

This course builds the model's collect → protect → analyze chain before the fleet arrives, so every later system is observed from its first boot. Collection is configuration you can show; protection is permissions and retention you can defend; analysis is a search someone actually runs.

Prerequisites

- ☐ Complete course 1.5: the admin plane exists, and this course's systems are managed only from VLAN 60.
- ☐ Provision SBD-SYSLOG01 (10.0.20.20) as Windows Server 2025, SBD-SPLUNK01 (10.0.50.10) on Linux joined to the sbd.test domain with its time service (chrony or ntpd) pointed at both domain controllers, and SBD-AUDIT01 (10.0.50.40) as Windows Server 2025 joined to the domain.
- ☐ Have the network's service-path register ready for the new rows: event forwarding to the collector, syslog from the core router, and the forwarder path into the analysis system.

What this course builds

- Forward the DC, admin-plane, and router-deny records into one collector by subscription.
- Protect the archive with hash catalogs, named readers, and delete-by-process-only.
- Prove the chain by tracing one event from source to analyst, with an owned alert.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-SYSLOG01 — name, address, OS build	
SBD-AUDIT01 — name, address, OS build	
SBD-SPLUNK01 — name, address, OS build	

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LESSON 01 / ABOUT 20 MIN

Design collect, protect, analyze

WHY THIS MATTERS Three systems, three distinct jobs: SBD-SYSLOG01 gathers, SBD-AUDIT01 preserves, SBD-SPLUNK01 answers questions. Decide what is collected first, how long it lives, and who may touch which stage — before a single event flows.

In this lesson you will

- Assign the three roles and the first collection scope.
- Decide retention, protection, and access per stage.
- Register the pipeline's network paths.

Follow the build

1. Write the role sentence for each host: the collector receives Windows events and the core router's syslog; the repository holds the protected, long-retention copy; the analysis system indexes a working copy for search and alerting. One copy to keep, one copy to question.
2. Scope the first collection to what already produces evidence: both DCs' Security logs, the core router's deny log, and the admin hosts' sign-in events — the fleet joins this scope in the next course, not now.
3. Decide the AU-family answers and write them down: retention length per stage, who may read each stage, and that nobody — including administrators — writes or deletes in the repository outside the archival process (that is the AU-9 protection story).
4. Add the register rows: DC and admin-host event forwarding to the collector (WinRM 5985), router syslog UDP 514 to the collector, and the forwarder path from collector to the analysis system; all three systems are managed only from VLAN 60 per course 1.5's handoff.

Check your result

```
# Capture the before state: what the DCs currently audit and where nothing is forwarded
yet.
auditpol /get /category:* | Select-String 'Logon|Account Management|Directory Service|
Policy Change'
Get-Service Wecsvc -ComputerName SBD-SYSLOG01
```

WHAT SUCCESS LOOKS LIKE The design names each stage's job, scope, retention, and readers; the repository's no-delete rule is written; and the register carries the three pipeline paths with owners.

Your record

Evidence to keep: Keep the pipeline design sheet, the retention and access table, and the new register rows.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

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RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why separate the repository from the analysis system?

REMEMBER Collector gathers, repository preserves, analysis answers. Scope starts with the DCs, the router denies, and the admin plane. Retention, readers, and the no-delete rule are written; paths are registered.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

NIST · SP 800-53 Rev. 5 security controls (AU family)

LESSON 02 / ABOUT 25 MIN

Turn on the right auditing

WHY THIS MATTERS Forwarding garbage faster is not monitoring. Set the advanced audit policy so the events that tell the network's story — sign-ins, account changes, directory changes, policy changes — actually get written.

In this lesson you will

- Apply the advanced audit policy categories the design needs.
- Reconcile audit settings with the course 1.3 Security Technical Implementation Guide (STIG) values.
- Verify effective auditing on the DCs and admin hosts.

Follow the build

1. Review what the Defense Information Systems Agency (DISA) Group Policy objects (GPOs) from course 1.3 already audit — much of the advanced audit policy arrived with the STIGs. This lesson reconciles and fills gaps rather than starting over.
2. Confirm the categories the design depends on are at least at the STIG values on the DCs: Logon/Logoff, Account Management, Directory Service Changes, and Policy Change — success and failure where the design says so.
3. Extend the same policy to the admin hosts' OU so privileged sessions write the sign-in trail lesson 5 of course 1.5 started capturing by hand.
4. Any place this course wants more auditing than the STIG sets, make it a documented addition in the GPO structure — never a local, invisible tweak — and note it in the exception/addition record.

Check your result

```
auditpol /get /subcategory:'Logon','User Account Management','Directory Service
Changes','Audit Policy Change'
gpresult /r /scope:computer
```

WHAT SUCCESS LOOKS LIKE Effective audit policy on the DCs and admin hosts covers the designed categories at or above the STIG values, applied through GPO, with any additions documented.

Your record

Evidence to keep: Keep the auditpol output per host class, the GPO report, and the documented additions.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

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RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Why start from the course 1.3 GPOs instead of a fresh audit policy?

> **REMEMBER** Audit policy is reconciled with the STIG baseline, not reinvented. DCs and admin hosts write the designed event families. Additions are GPO-visible and documented.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

Microsoft · Auditpol

LESSON 03 / ABOUT 30 MIN

Build the collector

WHY THIS MATTERS SBD-SYSLOG01 becomes the funnel: Windows Event Forwarding brings the DC and admin-host events in by subscription, and the core router's denies arrive as syslog — one host where the network's record first lands.

In this lesson you will

- Configure the Windows Event Collector service and subscription.
- Point the source hosts at the collector by GPO.
- Receive the router's syslog deny stream.

Follow the build

1. On SBD-SYSLOG01, enable the Windows Event Collector service and create a source-initiated subscription for the designed event IDs from the DCs and admin hosts, delivered into the Forwarded Events log.
2. By GPO on the relevant OUs, set the subscription manager to the collector and grant the Network Service read access to the Security log — the sources push; the collector never reaches into them.
3. Add a syslog intake on the collector for UDP 514 and point the core router's deny logging at it, so the network boundary's decisions land beside the Windows record.
4. Confirm the register rows are the only paths in use: WinRM from the source OUs to the collector, syslog from the core, and nothing listening that the design didn't ask for.

Check your result

```
wecutil es
wecutil gs SBD-ForwardedSecurity
Get-WinEvent -LogName ForwardedEvents -MaxEvents 5 | Select-Object TimeCreated, Id,
MachineName
```

WHAT SUCCESS LOOKS LIKE The subscription is active, events from both DCs and the admin hosts appear in Forwarded Events with correct source machine names, and router deny entries arrive on the syslog intake.

Your record

Evidence to keep: Record the subscription configuration, the GPO that enrolls sources, the first forwarded events per source, and a received router deny entry.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Why source-initiated subscriptions rather than the collector pulling?

> **REMEMBER** Sources push by subscription; the collector reaches into nothing. DC, admin-host, and router records land in one place. Only registered paths carry the pipeline.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

Microsoft · Wecutil (Windows Event Collector)

LESSON 04 / ABOUT 25 MIN

Protect the record on SBD-AUDIT01

> **WHY THIS MATTERS** The repository is where the record outlives mistakes and attackers alike: scheduled archives from the collector, hashed on arrival, readable by few, deletable by process only.

In this lesson you will

- Archive the collector's logs to the repository on schedule.
- Hash and catalog every archive on arrival.
- Enforce the read-few, delete-by-process-only access model.

Follow the build

1. Create the archival job: on schedule, the collector exports its Forwarded Events and syslog intake to dated archive files and transfers them to SBD-AUDIT01's repository share — the only inbound write path the repository allows.
2. On arrival, hash each archive and append the hash to a catalog kept with the archives; an archive whose hash no longer matches is a finding, not a file.
3. Set the permissions to match lesson 1's decision: the archival process writes, a named auditors group reads, and no interactive account holds delete rights — retention-driven deletion runs as its own documented, logged process, administered only under the audit custodian's separate accounts.
4. Fill in the AU table with what now exists: where each stage's data lives, its retention clock, and who can touch it — this table is the kind of artifact course 2.2 shows joining the Body of Evidence for the audit controls.

Check your result

```
Get-ChildItem \\SBD-AUDIT01\LogArchive | Select-Object Name, Length, LastWriteTime -First
5
Get-FileHash -Algorithm SHA256 \\SBD-AUDIT01\LogArchive\*.evtx | Select-Object -First 3
icacls \\SBD-AUDIT01\LogArchive
```

> **WHAT SUCCESS LOOKS LIKE** Dated archives arrive on schedule with catalog hashes that verify, the permission listing matches the access model exactly, and the AU table reflects reality.

Your record

Evidence to keep: Keep the archival job definition, the hash catalog excerpt, the permissions output, and the completed AU table.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

What does the hash catalog actually protect against?

REMEMBER Archives land on schedule and prove their own integrity. Readers are named; deletion is a process, not a right. The AU table is written for the authorization path.

Microsoft · Wecutil (Windows Event Collector)

NIST · SP 800-53 Rev. 5 security controls (AU family)

LESSON 05 / ABOUT 30 MIN

Analyze in SBD-SPLUNK01

> **WHY THIS MATTERS** Analysis earns its place with three working searches and one alert: failed privileged sign-ins, the boundary's denies, and the health of time — questions this network can already answer about itself.

In this lesson you will

- Index the collector's streams into the analysis system.
- Build the first three searches against real events.
- Create one alert with a named responder.

Follow the build

1. Install the analysis platform on SBD-SPLUNK01 and bring in the collector's two streams — the forwarded Windows events and the router syslog — over the registered forwarder path; the repository stays out of this flow by design.
2. Build search one: failed sign-ins (event 4625) involving the tier admin accounts, grouped by source host — the tiering course's denials now have a dashboard.
3. Build search two: router denies by source VLAN and destination port — the network boundary, now trending; and search three: time-service events and offsets, so clock drift surfaces before Kerberos complains.
4. Turn the first search into an alert with a threshold, a named responder from the admin plane, and a written first-response step — an alert nobody owns is a log entry with extra steps. Console access to all of this: VLAN 60 only, per the handoff.

Check your result

```
Test-NetConnection SBD-SPLUNK01.sbd.test -Port 9997
# In the analysis console, confirm the three saved searches return today's events
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4625} -MaxEvents 3 | Select-Object TimeCreated, MachineName
```

> **WHAT SUCCESS LOOKS LIKE** Both streams index with correct timestamps, the three searches return real current events, and the alert fires in a test with its responder and first step documented.

Your record

Evidence to keep: Record the input configuration, each search with a result sample, and the alert definition with its responder.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

Why does the repository stay out of the analysis flow?

> **REMEMBER** Two streams indexed; the protected record untouched. Three searches answer real questions from day one. One alert, one responder, one written first step.

Microsoft · Use Windows Event Forwarding to help with intrusion detection

Splunk · product documentation

LESSON 06 / ABOUT 20 MIN

Prove the chain and delta scan

> **WHY THIS MATTERS** One deliberate event, followed all the way: a failed privileged sign-in should appear on the source, the collector, the repository's next archive, and the analyst's screen — then the cadence scans close the course.

In this lesson you will

- Trace a test event through all three stages.
- Verify the router-deny stream the same way.
- Baseline the new hosts and file the course evidence.

Follow the build

1. Generate the test event: one failed sign-in on a DC with a wrong password for a tier admin account. Find it in the DC's Security log, in Forwarded Events on the collector, flag it for the next archive's verification on the repository, and watch the alert fire in analysis.
2. Repeat with the boundary: probe one unregistered port, then find that deny in the syslog stream and in search two — the boundary's evidence pair — attempt and deny — now assembles itself.
3. Run the cadence: first baselines for SBD-SYSLOG01, SBD-AUDIT01, and SBD-SPLUNK01 (Windows hosts with the standard benchmark; SBD-SPLUNK01 runs Linux, so scan it with the DISA STIG benchmark for its distribution), plus a DC rescan for the audit-policy deltas.
4. File the end-to-end trace as this course's signature evidence and write the handoff: the fleet course inherits a pipeline that will see its first domain join happen.

Check your result

```
Get-WinEvent -FilterHashtable @{LogName='ForwardedEvents'; Id=4625; StartTime=(Get-
Date).AddHours(-1)} | Select-Object TimeCreated, MachineName
Get-ChildItem C:\Evidence\logging-pipeline -Recurse | Select-Object Name, LastWriteTime
Get-FileHash -Algorithm SHA256 C:\Evidence\logging-pipeline\*
```

> **WHAT SUCCESS LOOKS LIKE** The test event appears at every stage with consistent timestamps, the deny trace matches, the new hosts have baselines, and the DC rescan deltas are attributed to the audit policy.

Your record

Evidence to keep: Keep the four-stage trace of the test event, the deny trace, the scan set, and the handoff note to course 1.7.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	

RECORD	YOUR OBSERVATION
Evidence file names	
Open items or findings	

Think it through

What does the single traced event prove that stage-by-stage checks don't?

> **REMEMBER** One event, four stages, consistent timestamps — the chain is real. Boundary denies assemble their own evidence now. Baselines, rescans, and the handoff to course 1.7 close the course.

Microsoft · Wecutil (Windows Event Collector)

NIST · SP 800-53 Rev. 5 security controls (AU family)

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Design collect, protect, analyze	☐ yes	
02 Turn on the right auditing	☐ yes	
03 Build the collector	☐ yes	
04 Protect the record on SBD-AUDIT01	☐ yes	
05 Analyze in SBD-SPLUNK01	☐ yes	
06 Prove the chain and delta scan	☐ yes	

Open items carried forward

Next in the learning path

That completes course 1.6. Course 1.7 brings in the fifty-workstation fleet — watched, addressed, and policy-managed from first boot.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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